Concrete de Rham Cohomology on S^2 and T^2, Winding and Area Forms, Euler Class, and Gauss–Bonnet

Contents

1 Goal **1**

2 de Rham cohomology **1**

3 Warm-up: the winding form on $\mathbb{R}^2 \setminus \{0\}$ **1**
3.1 Polar-coordinate computation . . . 2
3.2 Closed but not exact . . . 2

4 The sphere S^2: area form and cohomology **2**
4.1 Metric and area form . . . 2
4.2 Closed but not exact . . . 3
4.3 Why $H^1_{\mathrm{dR}}(S^2) = 0$ . . . 3

5 Local primitives on S^2 and overlap winding **3**

6 The torus $T^2 = \mathbb{R}^2/\mathbb{Z}^2$ **4**
6.1 Closed 1-forms dx, dy . . . 4
6.2 The top-degree class $dx \wedge dy$ . . . 4
6.3 Ring structure . . . 5

7 Local-to-global viewpoint: Čech cocycles and line bundles **5**

8 The Euler class and the curvature form **5**

9 Gauss–Bonnet on S^2 and T^2 **6**
9.1 Sphere . . . 6
9.2 Flat torus . . . 7
9.3 Important distinction . . . 7

10 Explicit stereographic-coordinate proof that the transition function has degree 2 **7**
10.1 Stereographic charts . . . 7
10.2 Overlap map . . . 8
10.3 Tangent transition function . . . 8
10.4 Connection with $\mathcal{O}(2)$ . . . 9

11 Big comparison **9**
11.1 Sphere S^2 . . . 9
11.2 Torus T^2 . . . 9

12 Deep geometric meaning **9**

1 Goal

We want a compute-first understanding of:

- de Rham cohomology through explicit differential forms,
- the winding form on $\mathbb{R}^2 \setminus \{0\}$,
- the area form on S^2,
- the basic forms $dx, dy, dx \wedge dy$ on T^2,
- the overlap/gluing viewpoint via Čech cocycles and transition functions,
- the Euler class of TS^2,
- and Gauss–Bonnet.

The central theme is:

$$\boxed{\text{closed} \neq \text{exact}}$$

and this failure is measured by topology.

2 de Rham cohomology

For a smooth manifold M,

$$H^k_{\mathrm{dR}}(M) = \frac{\{\text{closed } k\text{-forms}\}}{\{\text{exact } k\text{-forms}\}}.$$

So:

- ω is *closed* if $d\omega = 0$,
- ω is *exact* if $\omega = d\eta$,
- two closed forms define the same cohomology class if they differ by an exact form.

Thus a nonzero cohomology class is represented by a closed form which is not exact.

3 Warm-up: the winding form on $\mathbb{R}^2 \setminus \{0\}$

Consider

$$\eta = \frac{-y\,dx + x\,dy}{x^2 + y^2}.$$

3.1 Polar-coordinate computation

Let

$$x = r\cos\theta, \qquad y = r\sin\theta.$$

Then

$$dx = \cos\theta\,dr - r\sin\theta\,d\theta, \qquad dy = \sin\theta\,dr + r\cos\theta\,d\theta.$$

Hence

$$\begin{aligned} -y\,dx + x\,dy &= -r\sin\theta(\cos\theta\,dr - r\sin\theta\,d\theta) + r\cos\theta(\sin\theta\,dr + r\cos\theta\,d\theta) \\ &= -r\sin\theta\cos\theta\,dr + r^2\sin^2\theta\,d\theta + r\sin\theta\cos\theta\,dr + r^2\cos^2\theta\,d\theta \\ &= r^2\,d\theta. \end{aligned}$$

Since $x^2 + y^2 = r^2$, we get

$$\eta = d\theta.$$

So η is locally exact.

3.2 Closed but not exact

Since $\eta = d\theta$ locally,

$$d\eta = 0.$$

Thus η is closed on $\mathbb{R}^2 \setminus \{0\}$.

Now integrate around the unit circle $\gamma(t) = (\cos t, \sin t)$, $0 \le t \le 2\pi$:

$$\int_\gamma \eta = \int_0^{2\pi} d\theta = 2\pi.$$

If $\eta = df$ globally, then $\int_\gamma \eta = 0$, contradiction. So η is not exact.

Therefore

$$[\eta] \neq 0 \in H^1_{\mathrm{dR}}(\mathbb{R}^2 \setminus \{0\}).$$

Remark 1. *This is the prototype: there are local potentials, but no global single-valued potential.*

4 The sphere S^2: area form and cohomology

Let $S^2 \subset \mathbb{R}^3$ be the unit sphere. Use spherical coordinates

$$X(\theta, \phi) = (\sin\phi\cos\theta, \sin\phi\sin\theta, \cos\phi), \qquad 0 < \theta < 2\pi,\ 0 < \phi < \pi.$$

4.1 Metric and area form

A standard computation gives

$$ds^2 = d\phi^2 + \sin^2\phi\,d\theta^2.$$

Hence the area form is

$$dA = \sin\phi\,d\theta \wedge d\phi.$$

4.2 Closed but not exact

Since S^2 is 2-dimensional, every 3-form is zero, so

$$d(dA) = 0.$$

Thus dA is closed.

Now compute its integral:

$$\begin{aligned} \int_{S^2} dA &= \int_0^{2\pi}\int_0^{\pi} \sin\phi\,d\phi\,d\theta \\ &= \int_0^{2\pi} [-\cos\phi]_{\phi=0}^{\phi=\pi}\,d\theta \\ &= \int_0^{2\pi} 2\,d\theta \\ &= 4\pi. \end{aligned}$$

If $dA = d\alpha$ globally, then by Stokes,

$$\int_{S^2} dA = \int_{S^2} d\alpha = \int_{\partial S^2} \alpha = 0,$$

contradiction. So dA is not exact.

Therefore

$$[dA] \neq 0 \in H^2_{\mathrm{dR}}(S^2).$$

4.3 Why $H^1_{\mathrm{dR}}(S^2) = 0$

Concretely, every loop on S^2 bounds a region. So if ω is a closed 1-form and $\gamma = \partial D$, then

$$\int_\gamma \omega = \int_D d\omega = 0.$$

Thus all periods vanish, so there is no winding-type obstruction. Hence every closed 1-form is exact:

$$H^1_{\mathrm{dR}}(S^2) = 0.$$

So:

$$H^0_{\mathrm{dR}}(S^2) \cong \mathbb{R}, \qquad H^1_{\mathrm{dR}}(S^2) = 0, \qquad H^2_{\mathrm{dR}}(S^2) \cong \mathbb{R},$$

with $[dA]$ generating H^2.

5 Local primitives on S^2 and overlap winding

Cover S^2 by

$$U_N = S^2 \setminus \{N\}, \qquad U_S = S^2 \setminus \{S\}.$$

On U_N, define

$$\alpha_N = (1 - \cos\phi)\, d\theta.$$

Then

$$d\alpha_N = d(1 - \cos\phi) \wedge d\theta = \sin\phi\, d\phi \wedge d\theta = -\sin\phi\, d\theta \wedge d\phi.$$

So up to orientation convention,

$$d\alpha_N = dA.$$

On U_S, define

$$\alpha_S = -(1 + \cos\phi)\, d\theta.$$

Again,

$$d\alpha_S = dA$$

up to the same convention.

On the overlap $U_N \cap U_S$,

$$\alpha_N - \alpha_S = 2\, d\theta.$$

Thus the area form is locally exact on each chart, but the local primitives differ by the winding form $2\, d\theta$ on the overlap.

This is the concrete obstruction to global exactness.

6 The torus $T^2 = \mathbb{R}^2/\mathbb{Z}^2$

Let $T^2 = \mathbb{R}^2/\mathbb{Z}^2$, with coordinates (x, y) modulo integers.

6.1 Closed 1-forms dx, dy

The forms dx and dy descend to globally defined 1-forms on T^2, and

$$d(dx) = 0, \qquad d(dy) = 0.$$

So both are closed.

Now consider the loops

$$\gamma_x(t) = (t, 0) \mod \mathbb{Z}^2, \qquad \gamma_y(t) = (0, t) \mod \mathbb{Z}^2, \qquad 0 \le t \le 1.$$

Then

$$\int_{\gamma_x} dx = 1, \qquad \int_{\gamma_x} dy = 0,$$
$$\int_{\gamma_y} dx = 0, \qquad \int_{\gamma_y} dy = 1.$$

So neither dx nor dy is exact. Hence

$$[dx], [dy] \neq 0 \in H^1_{\mathrm{dR}}(T^2),$$

and they are linearly independent.

Therefore

$$H^1_{\mathrm{dR}}(T^2) \cong \mathbb{R}^2$$

with basis $[dx], [dy]$.

6.2 The top-degree class $dx \wedge dy$

Consider

$$dx \wedge dy.$$

It is closed because

$$d(dx \wedge dy) = 0.$$

Its integral is

$$\int_{T^2} dx \wedge dy = \int_0^1 \int_0^1 dx\, dy = 1.$$

So it cannot be exact.

Hence

$$[dx \wedge dy] \neq 0 \in H^2_{\mathrm{dR}}(T^2),$$

and it generates $H^2_{\mathrm{dR}}(T^2) \cong \mathbb{R}$.

Thus

$$H^0_{\mathrm{dR}}(T^2) \cong \mathbb{R}, \qquad H^1_{\mathrm{dR}}(T^2) \cong \mathbb{R}^2, \qquad H^2_{\mathrm{dR}}(T^2) \cong \mathbb{R}.$$

6.3 Ring structure

The wedge product gives

$$[dx] \wedge [dx] = 0, \qquad [dy] \wedge [dy] = 0,$$
$$[dx] \wedge [dy] = [dx \wedge dy], \qquad [dy] \wedge [dx] = -[dx \wedge dy].$$

So the top class on the torus is the wedge of the two basic winding classes.

7 Local-to-global viewpoint: Čech cocycles and line bundles

Suppose ω is a closed 2-form on a manifold M. By the Poincaré lemma, on each sufficiently small open set U_i we can write

$$\omega|_{U_i} = d\alpha_i.$$

On an overlap $U_i \cap U_j$,

$$d(\alpha_i - \alpha_j) = 0.$$

If the overlap is simply connected, then

$$\alpha_i - \alpha_j = df_{ij}$$

for some function f_{ij}.

So local primitives differ by exact terms on overlaps:

$$\alpha_i - \alpha_j = df_{ij}.$$

These f_{ij} measure the obstruction to gluing the local primitives α_i into one global 1-form.

Exponentiating gives transition functions

$$g_{ij} = e^{if_{ij}},$$

which satisfy cocycle conditions on triple overlaps and define a line bundle.

Remark 2. *This is the bridge:*

closed form $\longleftrightarrow$ *local primitives* $\longleftrightarrow$ *overlap functions* $\longleftrightarrow$ *transition functions of a bundle.*

8 The Euler class and the curvature form

Let M be an oriented Riemannian surface. On each sufficiently small open set U_i, choose a positively oriented orthonormal frame

$$(e_1^{(i)}, e_2^{(i)}).$$

The Levi-Civita connection is described by a local 1-form ω_i such that

$$\nabla e_1^{(i)} = \omega_i \, e_2^{(i)}, \qquad \nabla e_2^{(i)} = -\omega_i \, e_1^{(i)}.$$

The structure equation says

$$d\omega_i = K \, dA$$

on each chart. So the curvature form $K \, dA$ is locally exact.

On an overlap $U_i \cap U_j$, the two oriented orthonormal frames differ by a rotation:

$$\begin{pmatrix} e_1^{(j)} \\ e_2^{(j)} \end{pmatrix} = \begin{pmatrix} \cos\theta_{ij} & -\sin\theta_{ij} \\ \sin\theta_{ij} & \cos\theta_{ij} \end{pmatrix} \begin{pmatrix} e_1^{(i)} \\ e_2^{(i)} \end{pmatrix}.$$

Then

$$\omega_j = \omega_i + d\theta_{ij}.$$

So the connection 1-forms do not globally glue, but their differentials do:

$$d\omega_j = d\omega_i = K \, dA.$$

On triple overlaps,

$$\theta_{ij} + \theta_{jk} + \theta_{ki} \in 2\pi\mathbb{Z}.$$

Thus

$$n_{ijk} = \frac{1}{2\pi}\big(\theta_{ij} + \theta_{jk} + \theta_{ki}\big) \in \mathbb{Z}$$

defines an integer Čech 2-cocycle. This is the Euler class

$$e(TM) \in H^2(M;\mathbb{Z}).$$

Its de Rham image is

$$\boxed{\frac{1}{2\pi}[K\,dA] = e(TM) \in H^2_{\mathrm{dR}}(M).}$$

Integrating over M gives

$$\frac{1}{2\pi}\int_M K\,dA = \int_M e(TM) = \chi(M),$$

so

$$\boxed{\int_M K\,dA = 2\pi\chi(M).}$$

This is Gauss–Bonnet.

9 Gauss–Bonnet on S^2 and T^2

9.1 Sphere

For the unit sphere,

$$K \equiv 1.$$

So

$$K\,dA = dA,$$

and

$$\int_{S^2} K\,dA = \int_{S^2} dA = 4\pi.$$

Since

$$\chi(S^2) = 2,$$

we get

$$\int_{S^2} K\,dA = 2\pi\chi(S^2) = 4\pi.$$

Cohomologically,

$$[K\,dA] = [dA] \neq 0 \in H^2_{\mathrm{dR}}(S^2).$$

9.2 Flat torus

On the flat torus with metric

$$ds^2 = dx^2 + dy^2,$$

we have

$$K \equiv 0.$$

Hence

$$K\,dA = 0, \qquad [K\,dA] = 0 \in H^2_{\mathrm{dR}}(T^2),$$

and

$$\int_{T^2} K\, dA = 0.$$

Since

$$\chi(T^2) = 0,$$

Gauss–Bonnet gives

$$\int_{T^2} K\, dA = 2\pi\chi(T^2) = 0.$$

9.3 Important distinction

The area form dA on T^2 is not exact, because

$$\int_{T^2} dA = \mathrm{Area}(T^2) > 0.$$

So

$$[dA] \neq 0 \in H^2_{\mathrm{dR}}(T^2).$$

But for the flat torus,

$$K\, dA = 0.$$

So Gauss–Bonnet is about $[K\, dA]$, not about $[dA]$ by itself.

10 Explicit stereographic-coordinate proof that the transition function has degree 2

We now compute the tangent-bundle transition function of S^2 in stereographic coordinates.

10.1 Stereographic charts

Let

$$S^2 = \{(X, Y, Z) \in \mathbb{R}^3 : X^2 + Y^2 + Z^2 = 1\},$$

and let

$$N = (0, 0, 1), \qquad S = (0, 0, -1).$$

Define stereographic projection from the north pole:

$$\pi_N(X, Y, Z) = \frac{X + iY}{1 - Z}.$$

Define stereographic projection from the south pole:

$$\pi_S(X, Y, Z) = \frac{X - iY}{1 + Z}.$$

Write

$$z = \pi_N(p), \qquad w = \pi_S(p).$$

Remark 3. *The choice of $X - iY$ in the south chart makes the overlap map holomorphic in the oriented convention.*

10.2 Overlap map

For $p = (X, Y, Z) \in S^2 \setminus \{N, S\}$,

$$z = \frac{X + iY}{1 - Z}, \qquad w = \frac{X - iY}{1 + Z}.$$

Then

$$zw = \frac{(X + iY)(X - iY)}{(1 - Z)(1 + Z)} = \frac{X^2 + Y^2}{1 - Z^2}.$$

Since $X^2 + Y^2 + Z^2 = 1$,

$$X^2 + Y^2 = 1 - Z^2,$$

so

$$zw = 1.$$

Hence

$$w = \frac{1}{z}.$$

10.3 Tangent transition function

The tangent-bundle transition function is the derivative of the coordinate change:

$$g_{SN}(z) = \frac{dw}{dz}.$$

Since

$$w = \frac{1}{z},$$

we get

$$\frac{dw}{dz} = -\frac{1}{z^2}.$$

Now restrict to the equator $Z = 0$. There,

$$z = X + iY = e^{i\theta}.$$

So

$$g_{SN}(e^{i\theta}) = -e^{-2i\theta}.$$

This already shows the transition winds twice around S^1. Depending on whether one uses north-to-south or south-to-north clutching, and depending on the chosen orientation convention, one gets a degree $+2$ representative, usually recorded as

$$g(e^{i\theta}) = e^{2i\theta}.$$

The invariant geometric content is:

the tangent-bundle clutching map winds twice.

Therefore

$$e(TS^2)[S^2] = 2.$$

10.4 Connection with $\mathcal{O}(2)$

Identifying $S^2 \cong \mathbb{CP}^1$, the transition has degree 2, so as a complex line bundle,

$$TS^2 \cong \mathcal{O}(2).$$

Hence

$$c_1(TS^2) = 2.$$

For an oriented real rank-2 bundle underlying a complex line bundle,

$$e(TS^2) = c_1(TS^2) = 2.$$

Thus

$$\frac{1}{2\pi}\int_{S^2} K\,dA = 2, \qquad \int_{S^2} K\,dA = 4\pi.$$

11 Big comparison

11.1 Sphere S^2

$$H^0_{\mathrm{dR}}(S^2) \cong \mathbb{R}, \qquad H^1_{\mathrm{dR}}(S^2) = 0, \qquad H^2_{\mathrm{dR}}(S^2) \cong \mathbb{R}.$$

A generator of H^2 is $[dA]$. Since $K = 1$,

$$[K\,dA] = [dA] \neq 0.$$

Also,

$$\int_{S^2} K\,dA = 4\pi = 2\pi\chi(S^2).$$

11.2 Torus T^2

$$H^0_{\mathrm{dR}}(T^2) \cong \mathbb{R}, \qquad H^1_{\mathrm{dR}}(T^2) \cong \mathbb{R}^2, \qquad H^2_{\mathrm{dR}}(T^2) \cong \mathbb{R}.$$

Generators:

$$[dx], [dy] \in H^1, \qquad [dx \wedge dy] \in H^2.$$

For the flat torus,

$$[K\,dA] = 0, \qquad \int_{T^2} K\,dA = 0 = 2\pi\chi(T^2).$$

12 Deep geometric meaning

There are several equivalent ways to describe the same phenomenon:

- local potentials exist but global potentials may fail to exist;
- local frames exist but global frames may fail to exist;
- local connection forms exist but may fail to glue globally;
- the obstruction is recorded by cohomology, transition functions, and characteristic classes;
- curvature is the differential-form shadow of that obstruction.

For S^2, the obstruction is nontrivial:

$$e(TS^2) = 2, \qquad \chi(S^2) = 2, \qquad \int_{S^2} K\,dA = 4\pi.$$

For T^2, the curvature obstruction vanishes:

$$e(TT^2) = 0, \qquad \chi(T^2) = 0, \qquad \int_{T^2} K\,dA = 0.$$

13 Exercises

Exercise 1. *Show directly that*

$$d\left(\frac{-y\,dx + x\,dy}{x^2+y^2}\right) = 0 \quad \text{on } \mathbb{R}^2 \setminus \{0\}.$$

Exercise 2. *Compute*

$$\int_{|z|=1} \frac{-y\,dx + x\,dy}{x^2+y^2}.$$

Conclude that the winding form is not exact.

Exercise 3. *Verify that*

$$dA = \sin\phi\, d\theta \wedge d\phi$$

on the unit sphere.

Exercise 4. *Compute*

$$\int_{S^2} dA = 4\pi$$

and conclude that dA is not exact.

Exercise 5. *Compute*

$$d\big((1-\cos\phi)d\theta\big), \qquad d\big(-(1+\cos\phi)d\theta\big),$$

and compare the two local primitives on the overlap.

Exercise 6. *On $T^2 = \mathbb{R}^2/\mathbb{Z}^2$, compute the periods of dx and dy on the two basic cycles and prove that $[dx], [dy]$ are linearly independent.*

Exercise 7. *Compute*

$$\int_{T^2} dx \wedge dy$$

for the square torus and conclude that $dx \wedge dy$ is not exact.

Exercise 8. *Check Gauss–Bonnet directly for the unit sphere and for the flat torus.*

Exercise 9. *In the stereographic computation, explain carefully why the clutching map may appear as $-e^{-2i\theta}$ or $e^{2i\theta}$ depending on convention, while the geometric content remains "winding number 2".*